TENSOR-HOM ADJUNCTION FOR DIFFERENTIAL GRADED MODULES OVER A DGA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the right-hand identification is the defining universal property recorded in the same unit's tensor-product section (in-source reference); the left-hand unpacking uses the standard More on Algebra composition lemma for total complexes (navigation stub); both dependencies are stated in the proof and carried as stubs.

Editorial note. The label lemma-characterize-hom names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Universal property of the tensor product. Same unit, Section section-tensor-product. The proof identifies the right hand side with differential-compatible A-bilinear maps via the defining property of the tensor product of differential graded modules recorded in that section.
- Composition of total complexes. More on Algebra, Lemma more-algebra-lemma-compose (standard tooling). Left hand side: an element g of Hom(M', Hom(M, N)) determines a map of complexes g': Tot(M' tensor_R M) -> N by the composition lemma for total complexes.

Source reference. Original lemma 0A5Y: sources/stacks-chow/tags.txt, source lines 12826–12826. Prerequisite pointer invoked at proof line(s) 1782; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00CV: sources/stacks-dga/pinned-original-source.tex, source lines 1606–1606. Prerequisite pointer invoked at proof line(s) 1771; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let (A, d) be a differential graded R-algebra. Let M' be a right differential graded A-module and let M be a left differential graded A-module. Let $N^\bullet$ be a complex of R-modules. Then we have*

$$\mathrm{Hom}_{Mod_{(A,d)}}(M', \mathrm{Hom}(M, N^\bullet)) = \mathrm{Hom}_{Comp(R)}(M' \otimes_A M, N^\bullet)$$

where $M \otimes_A M$ is viewed as a complex of R-modules as in Section 00CV.

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Proof. Let us show that both sides correspond to graded A-bilinear maps

$$M' \times M \longrightarrow N^\bullet$$

compatible with differentials. We have seen this is true for the right hand side in Section 00CV. Given an element g of the left hand side, the equality of More on Algebra, Lemma 0A5Y determines a map of complexes of R-modules $g' : \mathrm{Tot}(M' \otimes_R M) \to N^\bullet$. In other words, we obtain a graded R-bilinear map $g'' : M' \times M \to N^\bullet$ compatible with differentials. The A-linearity of g translates immediately into A-bilinarity of g''. $\square$